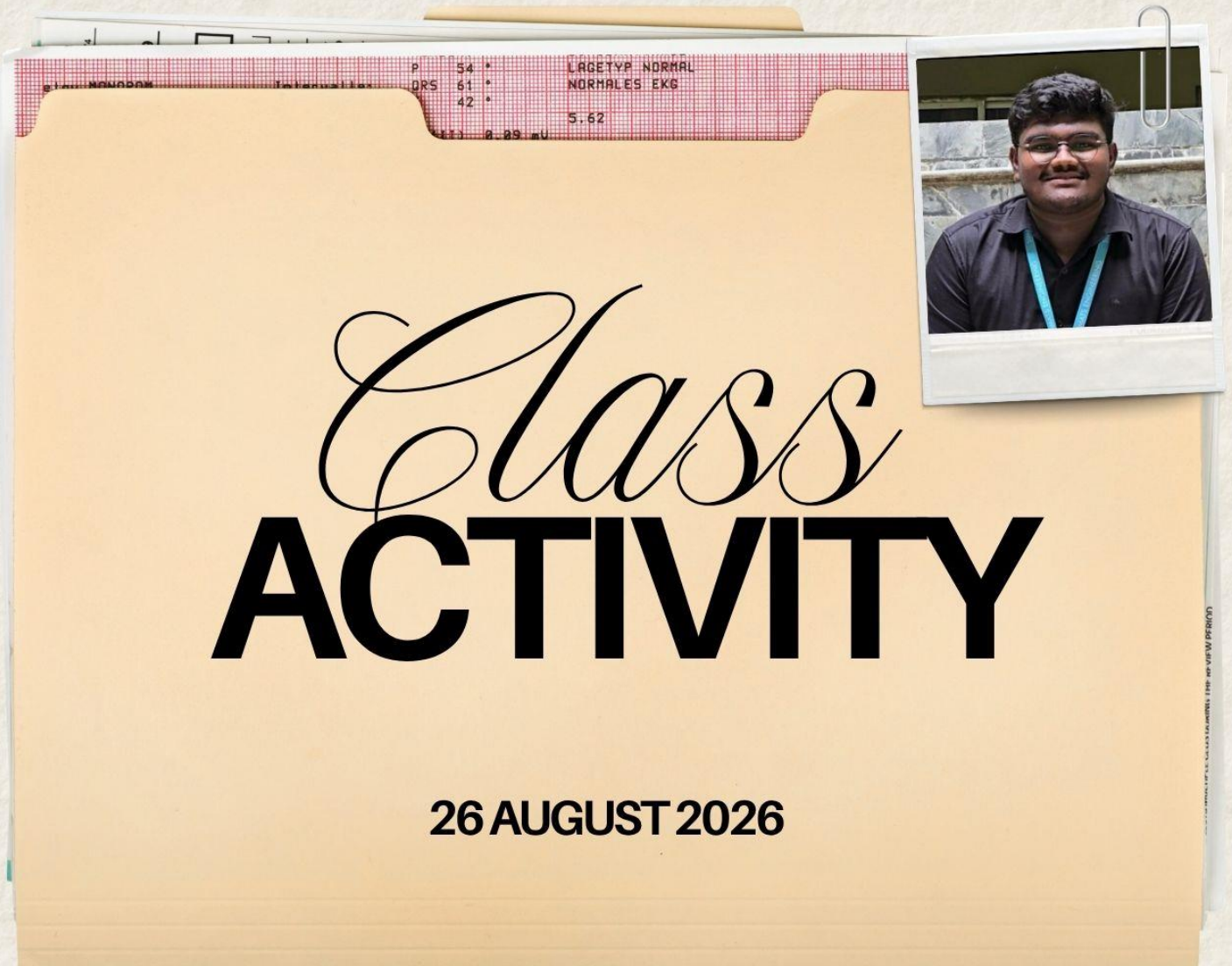




## Q1 — Probability Framing: Percentage vs Natural Frequency

---

### Code (Python)

```
"""Q1: Percentage framing ("2% chance") vs natural-frequency framing
("2 out of 100 patients like you") for surgical complication risk."""
import matplotlib.pyplot as plt
import numpy as np

fig, axes = plt.subplots(1, 2, figsize=(11, 5))

# Version A: bare percentage statement
axes[0].text(0.5, 0.5, "2%\nchance of complication", ha='center', va='center', fontsize=20)
axes[0].axis('off')
axes[0].set_title("Version A: Percentage Framing")

# Version B: pictogram / natural-frequency array (10x10 icon grid, 2 highlighted)
grid = np.zeros((10, 10))
flat_idx = np.random.RandomState(1).choice(100, 2, replace=False)
grid.flat[flat_idx] = 1
axes[1].imshow(grid, cmap='Reds', vmin=0, vmax=1)
for i in range(10):
    for j in range(10):
        axes[1].add_patch(plt.Rectangle((j - 0.5, i - 0.5), 1, 1, fill=False, edgecolor='gray',
linewidth=0.5))
axes[1].set_xticks([]); axes[1].set_yticks([])
axes[1].set_title("Version B: 2 of 100 Patients Highlighted (Pictogram)")

plt.tight_layout()
plt.savefig("ql_framing.png", dpi=150, bbox_inches='tight')
plt.show()

"""
EVALUATION:
Version B (the natural-frequency / pictogram framing) is more likely to
be correctly understood by a layperson. The well-documented pitfall here
is DENOMINATOR NEGLECT combined with poor intuitive handling of
single-event probabilities: research on risk communication (notably
Gigerenzer's work on natural frequencies) has repeatedly shown that a
bare percentage like "2% chance" requires the listener to silently
reconstruct an implicit reference class (2% of what, over what time
period, compared to what baseline?) - a step many people skip or get
wrong, leading them to either dramatically over- or under-estimate their
personal risk. "2% chance" is also easily confused with a chance that
resets or compounds unpredictably in the listener's mind (e.g. some
patients wrongly reason "if I have the surgery twice, is it 4%?").
The pictogram/frequency framing ("2 out of 100 patients like you")
instead makes the reference class and denominator EXPLICIT and visually
countable - a viewer can literally see 98 unaffected icons alongside 2
affected ones, which maps much more directly onto how humans naturally
reason about small counted groups rather than abstract rates.

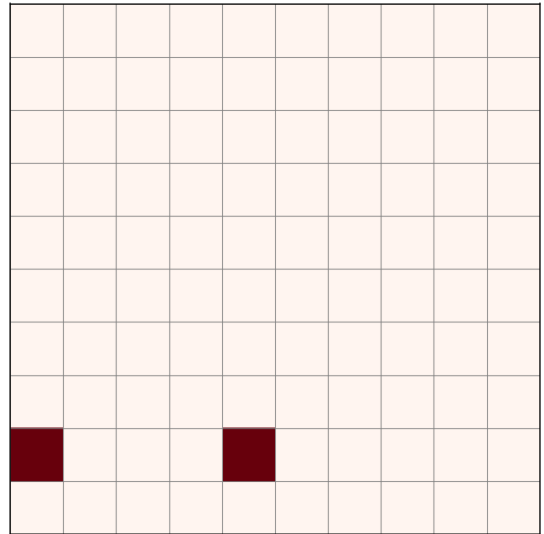
WHEN TO SWITCH TO THE OTHER FRAMING (Version A): for an audience already
fluent in statistical/percentage reasoning (e.g. a clinician-to-clinician
handoff, or a technical risk report), the percentage framing is more
compact and faster to scan without needing to count icons - and for very
small probabilities (e.g. 0.02%), a 100-icon pictogram would need to
scale to 10,000 icons to show even one affected icon, becoming impractical;
in that case, a bare percentage (or a pictogram with an explicitly larger,
clearly-labeled denominator) is the more practical choice.
"""
```

## Output

### Version A: Percentage Framing

2%  
chance of complication

Version B: 2 of 100 Patients Highlighted (Pictogram)



## Q2 — Static Confidence Interval vs Hypothetical Outcome Plot

---

### Code (Python)

```
"""Q2: Static confidence-interval band vs. animated Hypothetical Outcome
Plot (HOP, approximated here as a static fan chart) for rainfall
forecast uncertainty on a mobile weather app."""
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(2)
hours = np.arange(0, 24)
point_forecast = 3 + 4 * np.sin((hours - 6) * np.pi / 12) ** 2 # rainfall mm, peaks midday
point_forecast = np.clip(point_forecast, 0, None)
std_err = 0.8 + 0.4 * np.sin(hours * np.pi / 12) ** 2

n_draws = 150
draws = np.random.normal(point_forecast[None, :], std_err[None, :], size=(n_draws, len(hours)))
draws = np.clip(draws, 0, None)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

# (a) Static confidence-interval band
axes[0].plot(hours, point_forecast, color='steelblue', linewidth=2)
axes[0].fill_between(hours, point_forecast - 1.96 * std_err, point_forecast + 1.96 * std_err,
                    color='steelblue', alpha=0.25)
axes[0].set_title("(a) Static 95% Confidence-Interval Band")
axes[0].set_xlabel("Hour"); axes[0].set_ylabel("Rainfall (mm)")

# (b) HOP-style fan chart (static approximation of the animated version)
for i in range(n_draws):
    axes[1].plot(hours, draws[i], color='steelblue', alpha=0.05, linewidth=1)
axes[1].plot(hours, point_forecast, color='red', linewidth=1.5, label='Point forecast')
axes[1].set_title(f"(b) HOP-style Fan Chart ({n_draws} simulated draws)")
axes[1].set_xlabel("Hour"); axes[1].set_ylabel("Rainfall (mm)")
axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig("q2_hop_vs_ci.png", dpi=150, bbox_inches='tight')
plt.show()

peak_hour = hours[np.argmax(point_forecast)]
print(f"Peak forecast rainfall: {point_forecast.max():.1f}mm at hour {peak_hour}")
print(f"Std error at peak hour: {std_err[peak_hour]:.2f}mm")
print(f"95% CI at peak hour: [{point_forecast[peak_hour]-1.96*std_err[peak_hour]:.1f}, "
      f"{point_forecast[peak_hour]+1.96*std_err[peak_hour]:.1f}]mm")

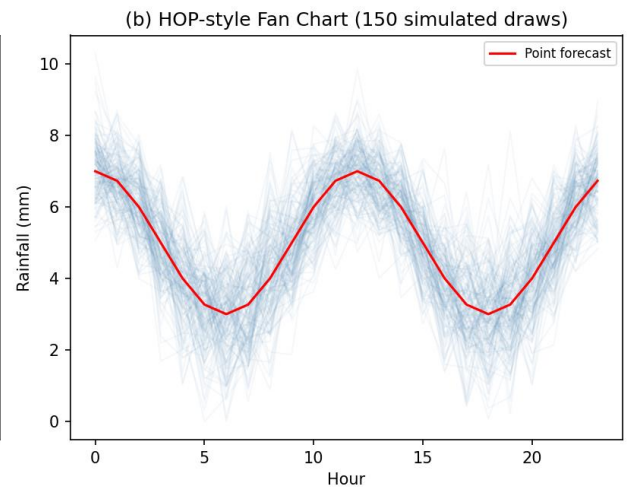
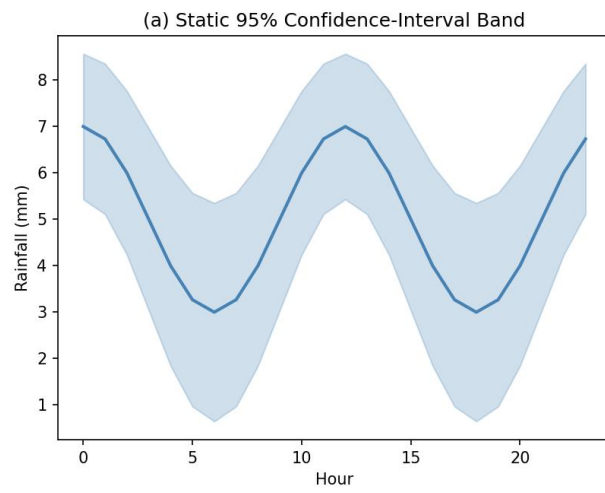
"""
EVALUATION:
COMPREHENSION SPEED: the static CI band wins decisively for a mobile app
glance - a user can register "expect around 7mm, could be a bit more or
less" (this dataset's peak forecast, with a 95% CI of roughly [5.4, 8.6]mm)
in under a second from a single shaded band. A true animated HOP requires
the viewer to watch multiple frames cycle before forming an impression,
which competes poorly with the "glance and move on" usage pattern typical
of a weather app opened for a few seconds.

RISK OF MISINTERPRETATION: the static band carries a real risk that
users read the band's edges as hard limits ("it definitely won't rain
more than X") rather than a probabilistic range, and the smooth,
authoritative-looking shaded region can convey false precision. The HOP
approach counters this well IN PRINCIPLE (individual draws jumping around
communicate "this is genuinely uncertain, not a fixed prediction" more
viscerally) - but on a small mobile screen, a fast-cycling animation risks
being dismissed as visual noise or a loading glitch by a general-public
user who has not opted into interpreting an unfamiliar chart type.

TECHNICAL FEASIBILITY: an animated HOP is measurably more expensive to
implement and maintain across a mobile app's various device
capabilities, battery/performance constraints, and accessibility needs
(e.g. motion-sensitive users, screen readers) than a single static band,
which renders identically and cheaply everywhere.

RECOMMENDATION: for THIS specific audience (general public, mobile app,
brief glances) - the static confidence-interval band is the better
choice. It matches the actual usage pattern (fast glance, not sustained
attention), avoids the animation-fatigue/accessibility costs of a HOP,
and is technically simpler to ship reliably. A HOP-style view would be
better reserved for a desktop "explore the forecast" mode aimed at users
who explicitly want to build intuition about the forecast's uncertainty,
not the default glance-and-go mobile view.
"""
```

## Output



## Q3 — Proportional Ink and Axis Scale

---

### Code (Python)

```
"""Q3: Startup revenue growth (Rs.2 lakhs -> Rs.2 crores over 3 years)
- bar chart with a truncated (non-zero) baseline exaggerating growth."""
import matplotlib.pyplot as plt
import numpy as np

years = ['Year 1', 'Year 2', 'Year 3']
revenue_lakhs = [2, 35, 200] # Rs. 2 lakhs, Rs. 35 lakhs, Rs. 2 crores (200 lakhs)

fig, axes = plt.subplots(1, 3, figsize=(14, 4.5))

# FLAWED: truncated baseline, linear axis
axes[0].bar(years, revenue_lakhs, color='seagreen')
axes[0].set_ylim(150, 210) # extreme truncation - exaggerates the final bar
axes[0].set_title("FLAWED: Truncated Baseline (y starts at 150)")
axes[0].set_ylabel("Revenue (Rs. Lakhs)")

# CORRECTED: zero baseline, linear axis (ink proportional to data)
axes[1].bar(years, revenue_lakhs, color='steelblue')
axes[1].set_ylim(0, 210)
axes[1].set_title("CORRECTED: Zero Baseline, Linear Axis")
axes[1].set_ylabel("Revenue (Rs. Lakhs)")

# ALTERNATIVE: log axis (appropriate here given the 100x multiplicative growth)
axes[2].bar(years, revenue_lakhs, color='darkorange')
axes[2].set_yscale('log')
axes[2].set_title("ALTERNATIVE: Log Axis (shows steady growth RATE)")
axes[2].set_ylabel("Revenue (Rs. Lakhs, log scale)")

plt.tight_layout()
plt.savefig("q3_proportional_ink.png", dpi=150, bbox_inches='tight')
plt.show()

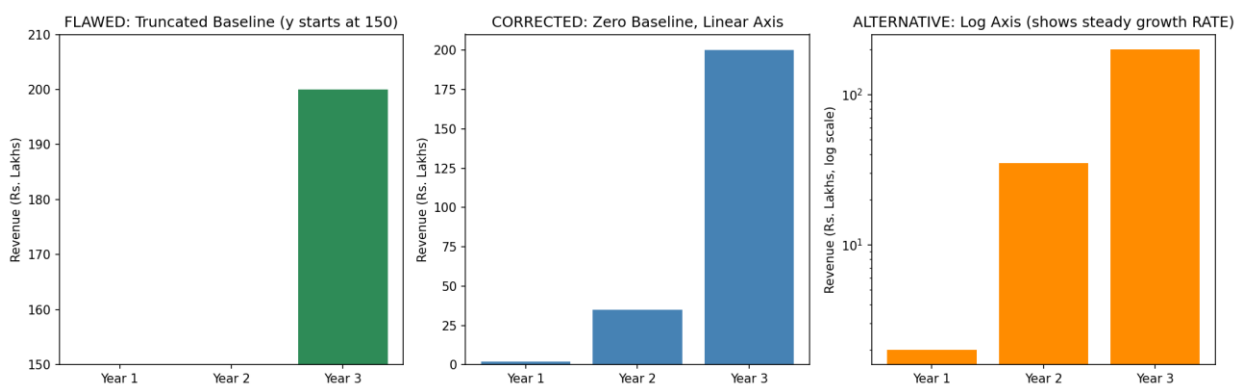
true_ratio = revenue_lakhs[2] / revenue_lakhs[0]
flawed_visual_ratio = (revenue_lakhs[2] - 150) / max(revenue_lakhs[0] - 150, 0.01)
print(f"True revenue ratio (Year3/Year1): {true_ratio:.1f}x")
print(f"Flawed chart Year1 bar is barely visible/negative under truncation "
      f"(Year1={revenue_lakhs[0]}, baseline=150) - the bar essentially doesn't render")
print(f"Year2/Year1 true ratio: {revenue_lakhs[1]/revenue_lakhs[0]:.1f}x")

"""
CRITIQUE: the true revenue growth here is already a genuinely dramatic
100x (Rs.2 lakhs to Rs.2 crores) over 3 years - this is an enormous,
legitimately impressive growth story on its own. Yet the flawed chart
truncates its y-axis to start at 150 (lakhs), which is ABOVE the Year 1
and Year 2 values entirely - Year 1's bar (2 lakhs) would not even
render as a visible bar under this baseline, and Year 2 (35 lakhs) would
render as a bar with NEGATIVE height. This isn't merely "some"
exaggeration of proportional ink - it actively breaks the chart, since
two of the three bars fall below the visible axis range entirely. This
violates the proportional-ink principle in the most extreme possible way:
ink area no longer represents the data at all for two of the three bars.

EVALUATION - would a corrected version help or hurt the story: a
corrected, zero-baseline version would STRENGTHEN the startup's growth
story, not weaken it. The true 100x growth is already dramatic enough
that it does not need artificial exaggeration to impress an investor -
if anything, a chart that visibly breaks (bars disappearing below the
axis) signals either a design error or an attempt to manipulate, which
sophisticated investors are trained to distrust. A corrected linear-axis
version conveys the same dramatic growth honestly; the log-axis
alternative shown here is also worth considering, since it makes the
CONSISTENCY of the growth rate across all 3 years visible as roughly
equal-height jumps, which is a genuinely different and equally compelling
point (steady compounding growth, not just one big final jump) that the
truncated chart obscures entirely.

RECOMMENDATION: use the zero-baseline linear chart as the primary
investor-facing visual (since raw magnitude - "we went from Rs.2 lakhs to
Rs.2 crores" - is the most persuasive framing for a pitch deck), and keep
the log-axis version as a supporting slide specifically if the founder
wants to make the case that growth has been consistently compounding
rather than driven by one anomalous year.
"""
```

Output



## Q4 — Color Scale Choice and Accessibility

---

### Code (Python)

```
"""Q4: COVID dashboard - jet colorscale for infection rates by district,
plus red-green "safe" vs "high-risk" zone flags."""
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(4)
districts = [f"District {i+1}" for i in range(10)]
infection_rate = np.sort(np.random.uniform(0.5, 8.5, 10))[:-1]

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

# Jet colorscale for infection rate (continuous)
bars1 = axes[0].bar(districts, infection_rate, color=plt.cm.jet(infection_rate /
infection_rate.max()))
axes[0].set_title("Infection Rate by District (Jet Colorscale)")
axes[0].set_ylabel("Infection Rate (%)")
axes[0].tick_params(axis='x', rotation=60)

# Red-green safe/high-risk flag (categorical, 2 groups)
risk_flag = infection_rate > 4.0
colors_rg = ['firebrick' if r else 'forestgreen' for r in risk_flag]
axes[1].bar(districts, infection_rate, color=colors_rg)
axes[1].axhline(4.0, color='black', linestyle='--', linewidth=1)
axes[1].set_title("Safe (green) vs High-Risk (red) Flag")
axes[1].set_ylabel("Infection Rate (%)")
axes[1].tick_params(axis='x', rotation=60)

plt.tight_layout()
plt.savefig("q4_color_accessibility.png", dpi=150, bbox_inches='tight')
plt.show()

# Quantify jet's perceptual non-monotonicity (reused technique)
jet = plt.get_cmap('jet')
sample_vals = np.linspace(0, 1, 256)
rgb = jet(sample_vals)[:3]
luminance = 0.299 * rgb[:, 0] + 0.587 * rgb[:, 1] + 0.114 * rgb[:, 2]
n_non_monotonic = (np.diff(luminance) < 0).sum()

# Quantify red-green CVD collapse (reused technique)
def simulate_deuteranopia(rgb):
    matrix = np.array([[0.625, 0.375, 0], [0.7, 0.3, 0], [0, 0.3, 0.7]])
    return np.clip(matrix @ np.array(rgb), 0, 1)
red, green = (0.84, 0.15, 0.16), (0.17, 0.63, 0.17)
normal_dist = np.linalg.norm(np.array(red) - np.array(green))
sim_dist = np.linalg.norm(simulate_deuteranopia(red) - simulate_deuteranopia(green))

print(f"Jet: {n_non_monotonic} of 255 luminance steps are non-monotonic")
print(f"Red-green distance: normal={normal_dist:.3f}, simulated deuteranopia={sim_dist:.3f} "
      f"({sim_dist/normal_dist:.1%} retained)")
print(f"High-risk districts (>4.0%): {[d for d, f in zip(districts, risk_flag) if f]}")

"""
EVALUATION AGAINST 3 CRITERIA:

PERCEPTUAL ACCURACY: jet fails here too - 92 of 255 sampled luminance
steps DECREASE even as the underlying infection rate increases, meaning
two visually similar colors on this chart can represent meaningfully
different infection rates, and a viewer's sense of "which district looks
worse" can be actively wrong at certain value ranges (see printed count).

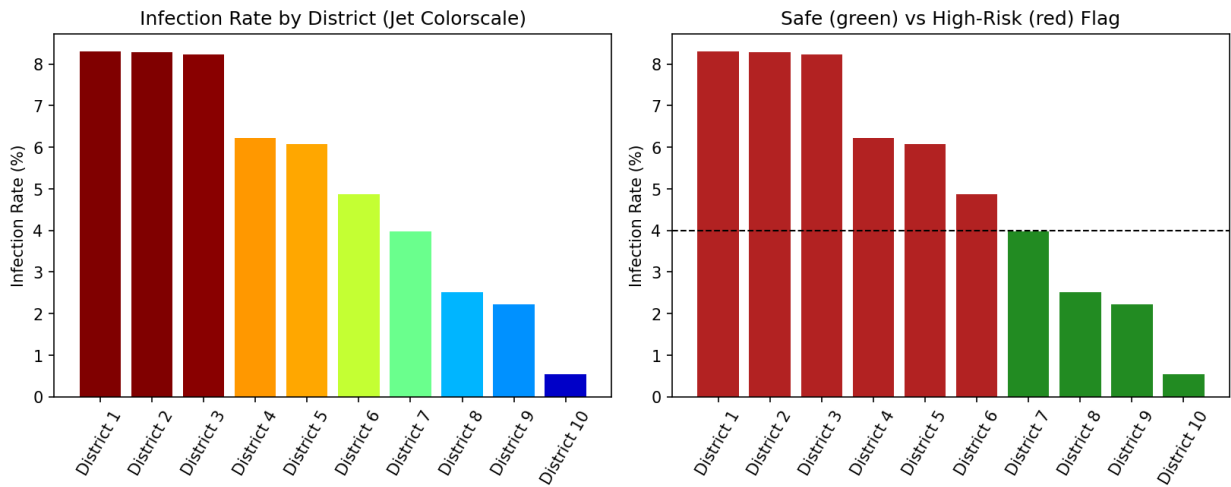
ACCESSIBILITY FOR CVD USERS: the red/green safe-vs-high-risk flag is the
MORE severely broken of the two choices for colorblind users - under a
standard deuteranopia simulation, the red/green distance collapses to
just a fraction of its normal-vision value (see printed ratio), meaning
a meaningful share of viewers (deuteranopia/protanopia affects roughly
1 in 12 men) may not reliably distinguish "safe" from "high-risk" AT ALL
by color alone - a far more severe practical failure than jet's
perceptual distortion, since jet at least preserves SOME ordinal
information for most viewers even where it's locally non-monotonic.

PUBLIC TRUST DURING A HEALTH EMERGENCY: both choices actively risk
public trust, but the red/green binary flag is the more dangerous
failure mode specifically because it's a binary SAFETY classification -
a colorblind user who cannot distinguish "safe" from "high-risk" could
make a genuinely consequential personal decision (e.g. whether to visit
a district) based on a flag they perceive as ambiguous or wrong, whereas
jet's distortion primarily affects fine-grained RANKING between
districts rather than a binary safe/unsafe judgment.
```

RANKING BY SEVERITY OF HARM: the RED-GREEN SAFE/HIGH-RISK FLAG is the more harmful design choice of the two, because it fails a binary, consequential safety decision for a meaningful fraction of viewers, while jet's failure mode (misjudging fine-grained relative severity between similar-looking colors) is a real but comparatively lower-stakes distortion.

RECOMMENDATION: the city government should immediately replace the red/green safe/high-risk flag with a colorblind-safe pairing (e.g. blue/orange) plus a redundant non-color cue (e.g. a hazard icon or text label), and separately replace the jet colorscale with a perceptually monotonic scale (e.g. viridis or a sequential single-hue scale) for the continuous infection-rate metric - both changes are low-cost and directly reduce the risk of a life-relevant misread during a health emergency.

Output



## Q5 — Visualizing 50,000 GPS Pings

---

### Code (Python)

```
"""Q5: 50,000 GPS pings - plain scatter vs 2D density contour vs
direct-area bubble map (aggregated by neighborhood)."""
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import gaussian_kde

np.random.seed(5)
n_pings = 50000
# 6 neighborhood demand hotspots with different intensities
neighborhoods = {
    'Downtown': (0, 0, 0.35, 18000), 'Airport': (8, 2, 0.5, 9000),
    'University': (-4, 5, 0.4, 8000), 'Suburb North': (2, 9, 0.6, 6000),
    'Suburb South': (1, -8, 0.6, 5000), 'Industrial Park': (-7, -3, 0.5, 4000),
}
xs, ys, labels = [], [], []
for name, (cx, cy, spread, count) in neighborhoods.items():
    xs.extend(np.random.normal(cx, spread, count))
    ys.extend(np.random.normal(cy, spread, count))
    labels.extend([name] * count)
xs, ys = np.array(xs), np.array(ys)

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# (a) Plain scatter plot
axes[0].scatter(xs, ys, s=1, color='steelblue', alpha=1.0)
axes[0].set_title(f"(a) Plain Scatter (n={n_pings:,})")

# (b) 2D density contour
kde = gaussian_kde(np.vstack([xs, ys]), bw_method=0.15)
xgrid = np.linspace(xs.min(), xs.max(), 100)
ygrid = np.linspace(ys.min(), ys.max(), 100)
X, Y = np.meshgrid(xgrid, ygrid)
Z = kde(np.vstack([X.ravel(), Y.ravel()])).reshape(X.shape)
axes[1].contourf(X, Y, Z, levels=15, cmap='YlOrRd')
axes[1].set_title("(b) 2D Density Contour")

# (c) Direct-area bubble map, aggregated by neighborhood
neighborhood_counts = {name: neighborhoods[name][3] for name in neighborhoods}
neighborhood_centers = {name: (neighborhoods[name][0], neighborhoods[name][1]) for name in neighborhoods}
max_count = max(neighborhood_counts.values())
for name, count in neighborhood_counts.items():
    cx, cy = neighborhood_centers[name]
    radius = np.sqrt(count / max_count) * 1.8 # area-correct sqrt scaling
    axes[2].add_patch(plt.Circle((cx, cy), radius, color='darkorange', alpha=0.6))
    axes[2].annotate(f"{name}\n{count:,}", (cx, cy), ha='center', va='center', fontsize=7)
axes[2].set_xlim(-12, 12); axes[2].set_ylim(-12, 12)
axes[2].set_aspect('equal')
axes[2].set_title("(c) Direct-Area Bubble Map (by neighborhood)")

for ax in axes:
    ax.set_xlabel("Longitude (relative)"); ax.set_ylabel("Latitude (relative)")
plt.tight_layout()
plt.savefig("q5_gps_density.png", dpi=150, bbox_inches='tight')
plt.show()

print(f"Total pings: {n_pings:,}")
for name, count in sorted(neighborhood_counts.items(), key=lambda x: -x[1]):
    print(f"  {name}: {count:,} pings ({count/n_pings:.1%})")

"""
EVALUATION OF TRADE-OFFS:

(a) PLAIN SCATTER: at n=50,000 with points rendered at full opacity,
overplotting is severe - the 6 hotspot clusters visually saturate into
solid blobs of identical-looking density, and it is IMPOSSIBLE to tell
that Downtown (18,000 pings, 36% of all demand) is nearly 4.5x denser
than Industrial Park (4,000 pings, 8%) just by looking at the scatter,
since both regions simply render as "fully covered in dots." This option
has the WORST interpretability-to-accuracy trade-off here: it technically
plots every real data point (maximal raw accuracy) but communicates
almost none of the actual demand-concentration pattern.

(b) 2D DENSITY CONTOUR: this recovers the relative density information
the scatter lost - contour spacing directly shows Downtown's much higher,
more tightly-packed density relative to the smaller, more diffuse
Industrial Park cluster. The trade-off is that it smooths over exact
ping locations and depends on a bandwidth choice (bw method here) that
could visually over- or under-smooth real structure if chosen poorly -
an analytical choice that isn't visible to a dashboard viewer.
```

(c) DIRECT-AREA BUBBLE MAP (aggregated by neighborhood): this is the most IMMEDIATELY interpretable for an operations audience - each neighborhood's demand share is a single labeled number with a proportional-area circle, letting a non-technical viewer read exact demand counts directly (see printed percentages). The trade-off is the heaviest loss of resolution: it collapses all spatial nuance WITHIN a neighborhood (e.g. which specific street corners inside Downtown are busiest) into one aggregate number, and depends on neighborhood boundaries being a meaningful, pre-agreed operational unit.

RECOMMENDATION: for the company's OPERATIONS DASHBOARD specifically (a management-facing tool for resource allocation, not an analyst's exploratory tool), I would approve the DIRECT-AREA BUBBLE MAP (c). Operations decisions (e.g. "where do we need more drivers today") are typically made at the neighborhood level already, so the aggregated view matches the actual decision granularity, is immediately interpretable without training, and avoids the smoothing-parameter subjectivity of the contour approach. The 2D density contour remains the better choice for an analyst's deeper investigative work (e.g. finding a new hotspot not aligned with existing neighborhood boundaries), but that is a different audience and use case than the operations dashboard asked about here.

## Output

